

Auditing GBFS bike-sharing feeds: a reproducible anomaly taxonomy for open mobility data, validated in France and projected across 48 countries

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Abstract

Open mobility feeds are increasingly assumed to be directly ingestible into quantitative research pipelines. The General Bikeshare Feed Specification (GBFS) is now mandatory for French operators and is catalogued worldwide by MobilityData, but syntactic standardisation does not guarantee the semantic viability that research reuse requires. We audit the 123 French GBFS systems and project the rule set across the 1,509-system MobilityData catalogue (48 countries), deriving a data-quality taxonomy of seven classes (A1–A7): five *structural errors* that violate the implicit semantic contract of the field they populate, and two *semantic warnings* for spec-compliant patterns that make the **capacity** column non-aggregable. Detection runs as a nine-step idempotent and reversible purging protocol, and we release the *GBFS Audit Catalogue* (46,307 stations, 97 cities, 46 typed columns) on Zenodo under ODbL. The protocol removes 30.9 % of the raw French stations and relabels a further 61 %, so fewer than 9 % survive into the catalogue untouched. The downstream effect is large: correcting a single class (A3) drops Bordeaux from the top quartile to below the median of the national dock-count distribution, which measures the policy cost of skipping the audit. The anti-patterns are global, not French-specific: 245 systems are flagged by A7 alone (97,314 stations), led by one operator that declares **capacity** = NaN across entire national deployments. Any quantitative use of GBFS data should therefore be preceded by an explicit and reproducible audit. The pipeline and a Docker image are released for reproducible rebuilds.

Highlights

- Seven-class taxonomy (5 structural, 2 semantic); projected across 1,509 GBFS systems
- Topology-aware A4 detector eliminates 8,005 centroid false positives
- Leave-one-operator-out CV on 7 operators supports rule generalisability
- 30.9 % of raw FR stations reclassified; hotspots are operator-driven
- Released as a versioned Parquet (46k stations) with Docker and a hold-out test

Keywords: GBFS; data quality; open mobility data; FAIR; reproducibility; bike sharing; standards compliance.

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1 Introduction

For more than a decade, bike-sharing has been a structural component of urban mobility policy in France and across Europe [1]. To support inter-operability between fleets, the General Bikeshare Feed Specification (GBFS), introduced by the North American Bikeshare Association in 2015 and now maintained by the Open Mobility Foundation, provides a shared ontology for station geolocation, dock counts, real-time availability and pricing [2, 3]. In France, the 2019 Mobility Orientation Law [4] (LOM, article L. 1115-1 of the transport code) makes publishing these feeds on the national portal *transport.data.gouv.fr* [5] a legal obligation. By 2026, many research and policy pipelines assume that GBFS feeds can be ingested directly.

That assumption is misleading. Our audit of 123 French GBFS systems shows that standardised syntax hides substantial semantic heterogeneity. The seven classes of Section 2.2 (Table 1) separate five *structural errors*, which violate the implicit semantic contract of the field they populate, from two *semantic warnings*, which flag spec-compliant patterns whose interpretation by a downstream consumer is ambiguous. The `station_information.capacity` field alone covers radically different realities: a physical dock count for dock-based fleets, an estimator obtained by conditional averaging on free-floating fleets, and arbitrary placeholders for under-reporting operators. The ambiguity reaches the object of study itself: seventeen car-sharing systems (fourteen under the *Citiz* brand) are listed on the national portal under the GBFS schema. It persists even among the systems that look clean, where 1.1 % of the certified stations are flagged by A4 as 3σ geospatial outliers (down from 3.6 % under the legacy centroid detector) and four further systems publish bounding boxes above the 50,000 km² A5 threshold or list stations outside the national perimeter. The methodological conclusion is straightforward: standardising the syntax of an open feed is not enough to make it a research artefact. Any quantitative use of GBFS data requires an explicit and reproducible audit.

Empirical audits of mobility feeds have so far remained piecemeal. Bike-sharing demand modelling and rebalancing analysis have generated a rich operational literature [6, 7, 8], but all of these works treat the input feeds as given and do not audit the standard itself. The closest precedent in transit-data quality is the smart-card literature: Pelletier et al. [9] systematically catalogue the gaps between what fare-collection feeds nominally publish and what they operationally support for research reuse, a programme close in spirit to the GBFS audit we report here. Tao et al. [10] document sampling biases in mobile-phone records used for origin-destination estimation; Romanillos et al. [11] warn against statistical artefacts of ingesting raw GPS feeds directly; Bachand-Marleau et al. [12] surfaced operational-definition heterogeneity in bike-sharing demand data more than a decade ago; and Fishman [13] stresses the heterogeneity of operational definitions in international BSS comparisons. The North American Bikeshare and Scootershare Association tracks cross-operator harmonisation gaps annually [14], with the regulatory framing typical of an industry-side body but without the per-row audit trail a research consumer needs. In the adjacent transit literature, the GTFS community has been more proactive: the Canonical GTFS Schedule Validator [15] and the earlier Conveyal validator [16] codify hundreds of schema-level checks on transit feeds, and the data-engineering community has produced reusable declarative-validation frameworks such as Great Expectations [17], Pandera [18] and the Frictionless Data specifications [19]. The broader anomaly-detection literature [20] provides a taxonomy of detection paradigms (statistical, proximity-based, clustering-based) that contextualises the spatial outlier methods we adopt in Section 4.2. The underlying data-quality literature is mature: Wang and Strong [21] established the consumer-centred dimensions, Pipino et al. [22] the assessment procedures, Khatri and Brown [23] the governance framing, Sebastian-Coleman [24] the ongoing-improvement perspective, and Naumann [25] the modern profiling approach. ISO/IEC 25012 [26] and its companion measurement standard ISO/IEC 25024 [27] formalise the quality dimensions; for geographic information specifically, ISO 19157 [28] provides the corresponding framework, and the European INSPIRE directive [29] makes them regulatory. W3C’s DCAT [30] and Data Quality Vocabulary [31] translate these dimensions into machine-

readable metadata. None of these references, however, addresses GBFS specifically, and none produces a versioned, audited reference dataset at the scale of a country. The present paper fills that gap for the French case and extends to the global GBFS catalogue.

This paper makes five contributions. First, a **taxonomy** of seven anomaly classes (A1–A7), derived from the audit of the 123 French GBFS systems and the 1,509-system MobilityData catalogue. It is stress-tested on a live panel of thirteen non-French systems spanning six station-management stacks (PBSC, JCDecaux Cyclocity, Smartbike, De Lijn / Blue-bike, Urban Sharing AS, Deutsche Bahn) across five countries; twelve serve as metropolitan negative controls and all pass classes A1, A2, A3, A6 and A7 under unchanged thresholds. Second, a nine-step **purging protocol** (Algorithm 1) that operationalises the seven classes and logs every transformation reproducibly and reversibly. Third, the **Audit Catalogue GBFS France** (46,307 certified stations, 97 cities), released as a versioned Parquet file with DOI 10.5281/zenodo.20125460 under ODbL, with its JSON Schema, Croissant manifest and audit report, and mirrored as a Hugging Face dataset. Fourth, an **open-source pipeline** (`audit_pipeline` Python package, MIT, ~ 560 LOC, unit-tested) and a five-tab **interactive dashboard** (<https://gbfs-audit.streamlit.app>) that exposes every audit verdict at the row level. Fifth, a **roadmap** of nine falsifiable experiments, six already with completed results, so that the protocol can be challenged in public rather than defended in private. Section 2 presents the taxonomy and the purging protocol, Section 3 reports the empirical impact on the French and global corpora, Section 4 the validation and robustness analyses, and Section 5 discusses implications and limitations.

2 Taxonomy and purging protocol

2.1 Provenance of the taxonomy

The seven-class taxonomy A1–A7 was built by a hybrid inductive-deductive workflow over six iterations between 2025-04 and 2026-04, starting from the four ISO/IEC 25012 data-quality dimensions (completeness, consistency, accuracy, currentness) and enriched by reverse engineering of both the French free-floating fleets and the global MobilityData catalogue. The five structural classes came first. A1 and A4 emerged from an exploratory scan of the 123 French systems, while A5 surfaced from enrichment with the *transport.data.gouv.fr* catalogue. Capacity diagnostics yielded A2 (a zero-variance signature on **capacity**) and A3 (a conditional-averaging estimator matching *Pony*’s per-system median capacity more closely than a physical-dock count). The two semantic warnings came later, from the global sweep: A6 from the empirical distribution of zero-capacity stations across the 1,509 globally audited systems (Section 2.3), and A7 from the bimodal 0%/100% distribution of NaN-capacity stations (Section 2.4). Two rejected hypotheses (“rolling-stock noise”, collapsed into A3; “cross-border duplicate”, no instance found) ship with the audit report for transparency.

2.2 The seven-class taxonomy

The audit of the 123 French GBFS systems, complemented by the global sweep of Section 3.4, defines seven recurring data-quality classes: five structural errors (A1–A5) and two semantic warnings (A6–A7), summarised in Table 1. Each class, in isolation, is sufficient to invalidate a non-audited cross-system comparison; the detection rules are formalised in Section 2.5.

The audit status of the 142 French GBFS feeds at the national portal is summarised in Figure 1: 123 systems are certified into the Audit Catalogue, 2 fail technical ingestion, and the 17 car-sharing systems (A1) are filtered upstream.

A3 is the class with the largest downstream impact on the French corpus. Free-floating fleets (*Pony*, *Cykleo*, etc.) advertise virtual stations in GBFS, the term “station” designating the GPS position of an anchoring point currently occupied by a vehicle. To avoid displaying empty stations, the operator typically reports a capacity profile by *conditional averaging* on stations

Table 1: The unified GBFS data-quality taxonomy: five *structural errors* (A1–A5) that violate the implicit semantic contract of the field they populate, plus two *semantic warnings* (A6–A7) that flag spec-compliant publication choices whose downstream-consumer interpretation is nevertheless ambiguous. Observed incidence on the released French catalogue (FR, 123 audited systems) and on the global MobilityData catalogue (Global, 1,509 systems across 48 countries). Each class is formalised in Section 2.5; per-system counts use the operational definitions documented in Section 2.5 (*e.g.* A3 flags any system with at least one `free_floating` entry after the S3 reclassification, which captures both natively-declared and post-audit-reclassified free-floating fleets). Counts are reproducible by re-running `python -m audit_pipeline.regenerate` on the released parquet.

Class	Name	Signature	FR	Global
<i>Structural errors (semantic-contract violations)</i>				
A1	Out-of-domain inclusion	Car-sharing advertised as BSS	17	46
A2	Placeholder capacity	Constant non-zero c_i across docked subset	1	48
A3	Structural over-capacity	Conditional averaging on free-floating	41	33
A4	Geospatial error	3σ outlier on per-system nearest-neighbour distance	88 (1.1 % stns)	81
A5	Out of perimeter	System bounding box $> 50,000 \text{ km}^2$	4	17
<i>Semantic warnings (spec-compliant but consumer-ambiguous)</i>				
A6	Zero-capacity dock	$\geq 1\%$ of docked stations declare $c = 0$	0	22
A7	Null capacity field	$\geq 50\%$ of stations declare $c = \text{NaN}$	32 (FF)	245

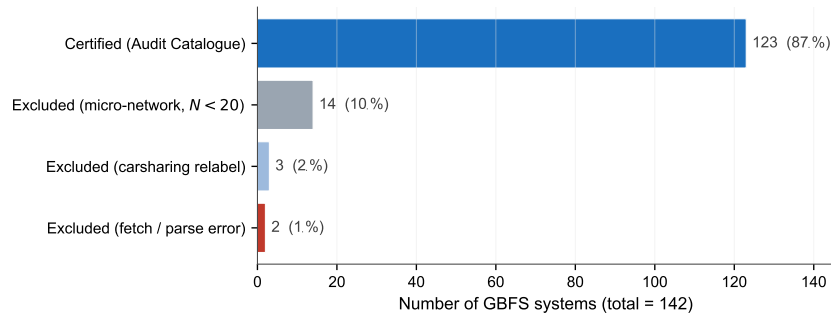


Figure 1: Verdict of the systematic audit applied to the 142 GBFS feeds inventoried on *transport.data.gouv.fr* and *MobilityData*.

whose instantaneous capacity is non-zero:

$$\bar{c}_{\text{profile}} = \frac{\sum_{i: c_i > 0} c_i}{\#\{i : c_i > 0\}} \neq \bar{c}_{\text{actual}} = \frac{1}{N} \sum_{i=1}^N c_i. \quad (1)$$

Aggregated at the system level, $\bar{c}_{\text{profile}}$ wrongly classifies thousands of free-floating bikes as heavy dock-based stations. Cross-system comparison is therefore meaningful only after explicit reclassification; the empirical signature is visible in Figure 2 as a bimodal capacity distribution.

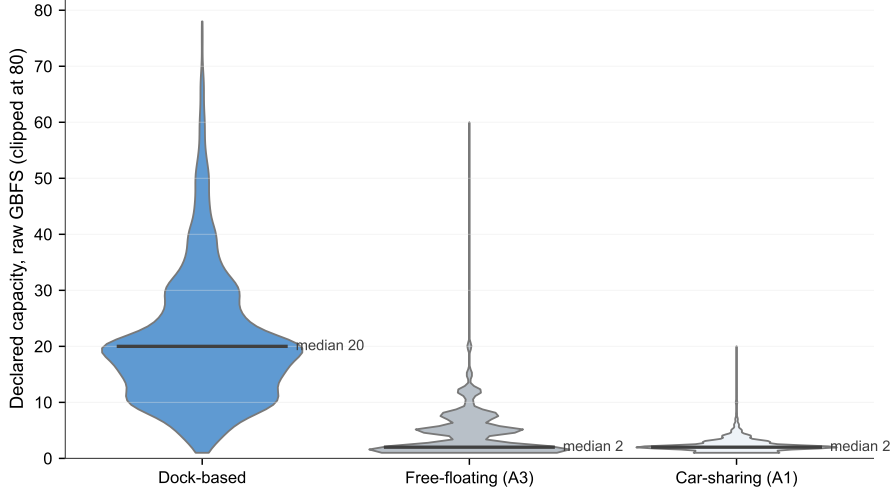


Figure 2: Empirical signature of the A3 floating-anchor bias. Declared capacity distribution by `station_type` (clipped at 80 docks) on the Audit Catalogue corpus. Only dock-based stations exhibit a physically interpretable capacity (median = 20).

2.3 A6: zero-capacity dock formalisation

The global scalability projection of Section 3.4 surfaced a recurring pattern not covered by A1–A5: a non-trivial fraction of stations in `station_information` declare `capacity = 0`. Because the GBFS schema reserves `station_information` for physical infrastructure and provides `station_status` for transient counts, a non-virtual physical station with zero docks is semantically inconsistent. The signature is therefore:

$$r_s^{A6} = \frac{\#\{i \in \mathcal{S}_s : c_i = 0 \wedge \text{is_virtual_station}(i) \neq \text{true}\}}{|\mathcal{S}_s|} \geq \tau_{A6}. \quad (2)$$

Threshold τ_{A6} and the empirical distribution. A6 is evaluated on the docked systems of the global sweep, where a zero-capacity station is semantically inconsistent; on free-floating fleets $c = 0$ is expected and carries no information, so they are excluded from the base. Across the 301 docked systems with at least 20 stations (Table S2), 89.0% declare *exactly zero* stations with $c = 0$, establishing the dock-based baseline, while the remaining 11.0% form a heavy tail with mean $\bar{r} = 0.36\%$ and standard deviation $\hat{\sigma} = 2.41\%$. The protocol threshold $\tau_{A6} = 1\%$ (Algorithm 1, step S7) isolates the upper tail: 22 docked systems globally cross it. The threshold is deliberately conservative, so that a flagged system declares zero-capacity on a non-trivial share of its docks rather than on the odd mis-entered station.

The empirical distribution of r_s^{A6} and the per-class incidence on the audited corpus are reported in Section 3.5 (Results, Table S2).

2.4 A7 (semantic warning): null capacity field formalisation

The Italian case study of Section 3.4 demonstrates that A1–A6 are structurally blind to operators that populate `station_information` but leave the `capacity` field unfilled. Like A6, A7 is by design a consumer-side semantic warning rather than a publisher-side anomaly. `capacity = NaN` is explicitly permitted by the GBFS v3 specification, in particular for free-floating fleets where the field has no physical referent; a system that publishes `capacity = NaN` on every entry is *spec-compliant*. What A7 captures is the downstream consequence: any analytical pipeline that sums or averages the `capacity` column across such a system aggregates NaN with non-NaN values silently, and produces a number whose physical meaning is undefined. The flag is therefore tooling for the consumer (“do not arithmetically combine these stations with others”), not a verdict on the publisher. The empirical signature is:

$$\rho_s^{A7} = \frac{\#\{i \in \mathcal{S}_s : c_i = \text{NaN}\}}{|\mathcal{S}_s|} \geq \tau_{A7}, \quad (3)$$

with the conservative threshold $\tau_{A7} = 0.5$ chosen so that a system flagged by A7 declares no usable capacity on at least half of its entries. Unlike A6, ρ_s^{A7} is extremely bimodal on the global corpus (Table S3): 87.0 % of audited systems sit at one of the two extreme values 0 % or 100 % NaN, with only 13.0 % distributed across the interior. Any threshold in $(0, 1)$ partitioning the interior in the same way yields essentially the same flagged set; $\tau_{A7} = 0.5$ is reported for parsimony and is robust to sensitivity sweeps in the tested range.

The empirical distribution of ρ_s^{A7} and the per-class incidence on the audited corpus are reported in Section 3.5 (Results, Table S3).

2.5 The nine-step purging protocol

The transition from the raw feed to the Audit Catalogue proceeds through a chain of nine sequential transformations (Algorithm 1), each coded in `utils/data_loader.py` and logged in an audit report shipped alongside the dataset. The protocol is deliberately sequential: early steps are inexpensive and remove out-of-domain or coarse-grained anomalies (A1, A2, A4 perimeter), while later steps require system-level statistics ($\bar{c}$, centroids, σ) and therefore benefit from operating on a pre-cleaned set.

2.6 Protocol properties

Four properties are designed into the pipeline. *Idempotence*: two successive applications produce the same result, tested by a unit test executed on every CI run. *Reversibility*: each excluded station is kept in `rejected_stations.parquet` together with the exclusion motive. *Logging*: every step records the number of stations in, the number of stations out and the type of transformation applied. *Explicit parameterisation*: all thresholds ($\sigma_{\max}$, $N_{\min}$, geofilter bounds, hybridisation radii) are documented as hyperparameters and can be varied for sensitivity analysis without modifying the source code.

These four properties are not arbitrary engineering choices; they operationalise the data-quality dimensions formalised by ISO/IEC 25012 [26] and ISO/IEC 25024 [27], and together they make the pipeline a direct domain-level implementation of the standard. Idempotence enforces the *internal consistency* dimension (two computations on the same input must yield the same output). Reversibility, through the `rejected_stations.parquet` sidecar and the systematic preservation of the exclusion motive, materialises the *recoverability* property and contributes to *traceability*. Logging at every step extends that traceability to the full pipeline run. Explicit hyperparameter declaration addresses the *understandability* dimension at the protocol level. Finally, the FAIR release of the audit report alongside DCAT-AP, JSON Schema and Croissant manifests (Section 2.10) closes the *compliance* requirement of 25012. Table 2 summarises the

Input: National catalogue $\mathcal{C}$ (123 systems); thresholds $\sigma_{\max} = 3$, $N_{\min} = 20$, $\tau_{A2}^{\text{size}} = 20$, $\tau_{A6} = 0.01$, $\tau_{A7} = 0.5$, perimeter $\Phi = [41^\circ, 52^\circ] \times [-6^\circ, 10^\circ]$.

Output: Certified set $\mathcal{G}$, rejected set $\mathcal{R}$, seven per-row anomaly flags $f_{A1} \dots f_{A7}$, audit report.

$\mathcal{G} \leftarrow \emptyset$; $\mathcal{R} \leftarrow \emptyset$

foreach *system* $s \in \mathcal{C}$ **do**

Ingest GBFS feed $\rightarrow \mathcal{S}_s$

S1. If s is car-sharing, relabel **carsharing** and set $f_{A1} = \text{True}$

S2. If $\text{Var}(c) = 0$, $c > 0$ and $|\mathcal{S}_s^{\text{dock}}| \geq \tau_{A2}^{\text{size}}$, set $f_{A2} = \text{True}$ and force **free_floating**

S3. Compute $\bar{c}_{\text{profile}}/\bar{c}_{\text{actual}}$; if > 5.0 relabel **free_floating** and set $f_{A3} = \text{True}$

S4. Keep $\mathbf{x}_i \in \Phi$; mark stations outside as out-of-perimeter

S5. Compute composite anomaly score $a_i = \alpha \hat{o}_i + (1 - \alpha) s_i$; if $a_i > 0.8$ set

$f_{A4} = \text{True}$ (Section 4.2); fallback: nearest-neighbour robust 3σ envelope

S6. Compute system bbox area A_s ; if $A_s > 50,000 \text{ km}^2$, set $f_{A5} = \text{True}$ on every $i \in \mathcal{S}_s$

S7. If $\#\{i : c_i = 0 \wedge \text{docked}\}/|\mathcal{S}_s| \geq \tau_{A6}$, set $f_{A6} = \text{True}$ on every $i \in \mathcal{S}_s$

S8. If $\#\{i : c_i = \text{NaN}\}/|\mathcal{S}_s| \geq \tau_{A7}$, set $f_{A7} = \text{True}$ on every $i \in \mathcal{S}_s$

S9. If $|\mathcal{S}_s^{\text{dock}}| < N_{\min}$, label **too_small** and route to $\mathcal{R}$

Update $\mathcal{G}, \mathcal{R}$ and the per-row flag matrix accordingly

end

return $(\mathcal{G}, \mathcal{R}, \{f_{A1} \dots f_{A7}\}, \text{audit report}, \text{metadata})$

Algorithm 1: GBFS raw-to-Audit-Catalogue purging pipeline. Steps S1–S3 both purge and label (a station relabelled to **free_floating** or **carsharing** stays in the catalogue with its A-flags set); S4 implements the national perimeter filter; S5 the A4 topology-aware outlier check (Section 4.2); S6 the A5 macro-region surface check; S7–S8 the A6 / A7 capacity-field warnings. S9 is the only truly-removing step.

mapping; we are not aware of a published reference implementation of ISO/IEC 25012 on an open mobility standard prior to the present work.

Table 2: Mapping of the nine-step purging-protocol design properties to the data-quality dimensions of ISO/IEC 25012:2008 and its companion measurement standard ISO/IEC 25024:2015. The Audit Catalogue is, to our knowledge, the first FAIR-released implementation of these dimensions on a GBFS corpus.

Protocol property	ISO/IEC 25012 dimension	Mechanism in the pipeline
Idempotence	Internal consistency	Unit-tested re-application on certified output
Reversibility	Recoverability + Traceability	<code>rejected_stations.parquet</code> with exclusion motivation
Step-level logging	Traceability	Per-step in/out counts and transformation type
Explicit parameterisation	Understandability	Hyperparameters in code + documented in audit report
FAIR release	Compliance + Accessibility	JSON Schema, DCAT-AP, Croissant, ODbL on Zenodo
Blind author cross-annotation	Accuracy + Semantic consistency	Author-led consistency check that flagged anomalies

2.7 Threshold justification

The topological threshold $\sigma_{\max} = 3$ is a compromise between robustness to GPS noise and the preservation of legitimate peripheral extensions, *e.g.* an off-centre TGV station or a university campus. Section 4.4 reports the single-dimension sensitivity pilot: Top-10 invariance for $\sigma_{\max} \geq 2.5$. The robustness threshold $N_{\min} = 20$ dock-based stations is calibrated on the literature of meshed networks: below this value, density and connectivity metrics become noise-dominated and are no longer informative for cross-city comparison [32]. The A5 macro-region threshold of 50,000 km² is calibrated at the order of magnitude of a French *métropolitaine région administrative* (median area $\approx 47,000$ km² after the 2016 reform: Bretagne 27,000, Pays de la Loire 32,000, Centre-Val de Loire 39,000, Hauts-de-France 31,000, Grand Est 57,000, Auvergne-Rhône-Alpes 69,000, Occitanie 72,000). A single GBFS feed whose bounding box exceeds this scale plausibly aggregates operationally-independent deployments and the resulting system-level statistics (centroid, intra-system KNN, capacity profile) are no longer geographically interpretable.

2.8 Multi-source hybridisation

The output of the purging stage enriches each station with a contextual profile obtained by articulating six reference sources (Table S5). Hybridisation is performed either by spatial joining (point-in-polygon, buffer queries) or by administrative aggregation (*i.e.* at the IRIS or municipal level), depending on the granularity of the source. Each row of Table S5 corresponds to one or more typed columns of the released parquet schema (Table S4).

2.9 Released schema

The Audit Catalogue Parquet exposes 46 columns documented in Table S4: 14 from the raw GBFS payload (preserved verbatim or normalised), 11 from the audit pipeline (typed `station_type` enum, capacity-raw and capacity-audited pair, seven per-station anomaly flags A1–A7, operator label, and audit-confidence rating), 5 from intra- and inter-system network geometry, 4 from spatial-equity hybridisation, 3 from infrastructure hybridisation, and 2 from multimodal-access hybridisation. Every column carries a stable name, a declared dtype, a source pointer and a completeness rate; the machine-readable equivalent is shipped as a JSON Schema, a Frictionless Data Package descriptor and a Croissant manifest. The eleven audit-pipeline columns are the key novelty of the catalogue: they make the audit’s verdict inspectable per row, rather than aggregated in a separate report.

Empirical completeness on the 5,442 dock-based stations is above 93 % for every contextual variable except the optional **address** field of the raw GBFS payload (Figure 3). The minimum-completeness threshold of 90 % documented in the audit report is therefore reached on every certified column of the released schema.

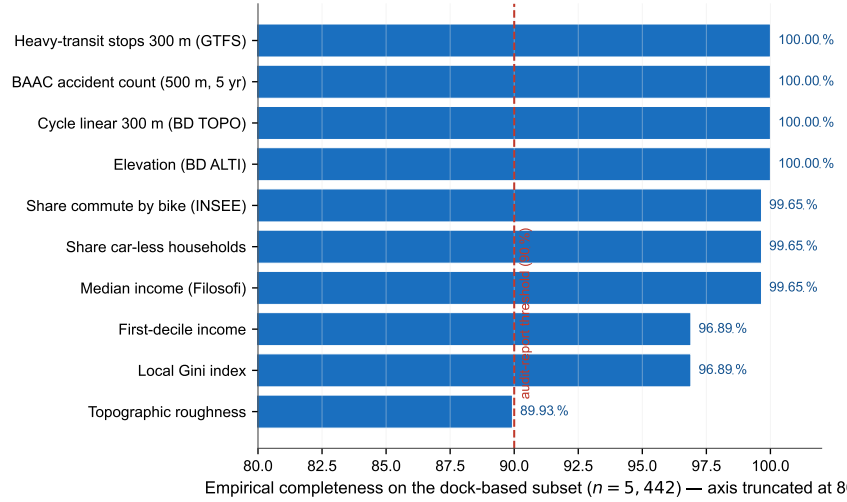


Figure 3: Empirical completeness of contextual enrichment per derived variable on the dock-based subset of the Audit Catalogue ($n = 5,442$). All variables exceed the 90 % completeness threshold required by the audit report; missing values remain below 6 % across the corpus.

2.10 Reproducibility infrastructure

The Audit Catalogue is released in alignment with the four FAIR principles [33]. *Findable*: primary deposit on Zenodo under concept DOI 10.5281/zenodo.20125460, with one DOI per SemVer-tagged release (current 1.0.0, dated 2026-05); mirrored on *Recherche Data Gouv* (HAL/BSO indexing). *Accessible*: Parquet format, direct HTTPS download and Git LFS mirror. *Interoperable*: the schema extends GBFS v3 and is published as a JSON Schema, a Frictionless Data Package descriptor [19], a DCAT-AP record [34] and a Croissant manifest [35]. *Reusable*: released under ODbL v1.0; code under MIT, with audit report and source code; the human-validation gold standard (`gold_labels.csv`), the inter-rater reliability report and the blind annotation tool ship with the repository (`experiments/annotation/`). Raw sources are archived with SHA-256 hashes; the Python environment is declared in `requirements.txt`; a Docker image is published for reproducible rebuilds. The audit pipeline itself is fully deterministic (no random sampling); the only stochastic step is the bootstrap resampling used for the confidence intervals reported in the main text, for which the random seed is pinned at `numpy.random.default_rng(42)`. Idempotence and per-class detector unit tests run in CI on every push; the audit report is regenerated automatically on every release tag.

2.11 Implementation

The released Parquet file is 7.8 MB on disk for the 46,307 stations and loads in under one second into a standard `pandas` session on commodity hardware; typical analytical queries (city-level filtering, bounding-box selection, joins with external panels) complete in tens of milliseconds. The audit pipeline is implemented as a compact Python package (`audit_pipeline`, ~ 560 lines of code across three modules: `core` for the Tier-1 / Tier-2 detectors, `regenerate` for one-command artefact refresh, and the package `__init__` for the public API) with stable function-level entry points documented in the source repository.

Software validation. The audit primitives are accompanied by a unit-test suite (`tests/test_core.py`, 24 tests, 85 % line coverage of the public `audit_pipeline.core` surface) that exercises each of the seven anomaly classes in isolation as well as the combined end-to-end `enrich()` pipeline. Each detector is paired with a deterministic synthetic fixture engineered to trip exactly that class (e.g., a 25-station grid with one transposed-coordinate row for A4, a 32-station fleet spanning more than 300,000 km² for A5, a 60 %-NaN-capacity dock-based system for A7) and with specificity controls verifying that healthy fixtures stay unflagged. The suite further checks that the Tier-2 geometry columns (intra- and inter-system k -nearest-neighbour distances, local densities) are consistent with the equirectangular projection used for the meter-scale arithmetic, and that the pipeline is idempotent under repeated application. Tests are executed by `pytest` as a GitHub Actions workflow on every push and pull request, and locally inside the published Docker image. This suite validates the correspondence between the implemented logic and the published anomaly taxonomy; it is not a substitute for the hand-curated cross-operator audit and the author-led consistency check (Section 4.5) against which the pipeline’s verdicts were calibrated.

3 Empirical results

3.1 Cumulative effect on the national corpus

Table 3 summarises the cumulative effect of the purging protocol. From 67,000 raw `station_information` entries published by the 142 inventoried French GBFS feeds, 20,693 (30.9 %) are removed from the catalogue as falling outside the certification criteria (excluded micro-networks below $N_{\min}$, technical-ingestion failures, perimeter and outlier filters); the remaining 46,307 stations are kept in the catalogue with an explicit `station_type` and per-row anomaly flags, of which a further 40,865 (61.0 % of the raw) are *relabelled* away from their declared GBFS type (39,235 relabelled to `free_floating` by step S3, 1,630 to `carsharing` by step S1). We report two complementary uncertainty bounds on the 30.9 % removal rate to surface the role of clustering. A *station-level* Wilson 95 % interval, ignoring within-system correlation, gives [30.54 %, 31.24 %]; this is the bound a naive independent-Bernoulli assumption would place on the figure and is a *lower bound* on the real uncertainty. A *size-stratified cluster bootstrap* (123 systems partitioned into size quartiles, resampled with replacement within each stratum, $n = 10,000$, seed 42) gives a 95 % interval of [−18.6 %, 63.6 %]; this is an *upper bound* because the largest size quartile contains a single dominant deployment (*Vélib’ Métropole*, $\sim 14,700$ stations) whose inclusion or exclusion across resamples mechanically shifts the denominator by an order of magnitude. The asymmetry of these two intervals, one tight and one degenerate, is itself the substantive observation: on a 123-system corpus with size spanning four orders of magnitude, the within-snapshot uncertainty on a stations-level total is dominated by which operators happen to be present, not by sampling noise within operators. We therefore report the 30.9 % removal rate as a deterministic accounting fact on the audit-dated snapshot, and defer temporal uncertainty to experiment E3 (twelve-month stability) where re-running the audit on freshly fetched snapshots gives a properly aligned variance estimate. The headline figure is that fewer than 9 % of raw entries survive into the catalogue without either being relabelled or removed, a strong empirical argument against the view that the standard alone delivers a research-ready dataset.

The most heavily impacted urban areas are those operated by free-floating fleets (*e.g.* Bordeaux, Nice, Saint-Étienne) and those historically lumped together with car-sharing systems on the national portal (Strasbourg, Grenoble), whereas urban areas served by traditional dock-based operators (*Vélib’* in Paris, *VéloMag* in Montpellier) are left almost intact. The certified-station mass by administrative region after purging is shown in Figure 4: Île-de-France concentrates the historical *Vélib’* deployment, while regions with many free-floating systems (Nouvelle-Aquitaine, Auvergne-Rhône-Alpes) hold a much smaller dock-based stock after the A3 correction is applied.

Table 3: Effect of the purging pipeline on the national GBFS corpus. *Certified* entries are kept in the Parquet with a typed `station_type` so that downstream consumers can filter explicitly; *rejected* entries are kept in a separate `rejected_stations.parquet` with the exclusion motive.

Indicator	Raw	Audit Catalogue
Inventoried GBFS systems	142	123 (certified)
Cities covered	—	97
with at least one dock-based station	—	64
Total stations (certified)	67,000	46,307
dock-based	—	5,442
free-floating (labelled, preserved)	—	39,235
car-sharing (A1)	1,630	re-labelled, excluded from BSS use
Stations dropped from catalogue	—	20,693 (30.9 %)

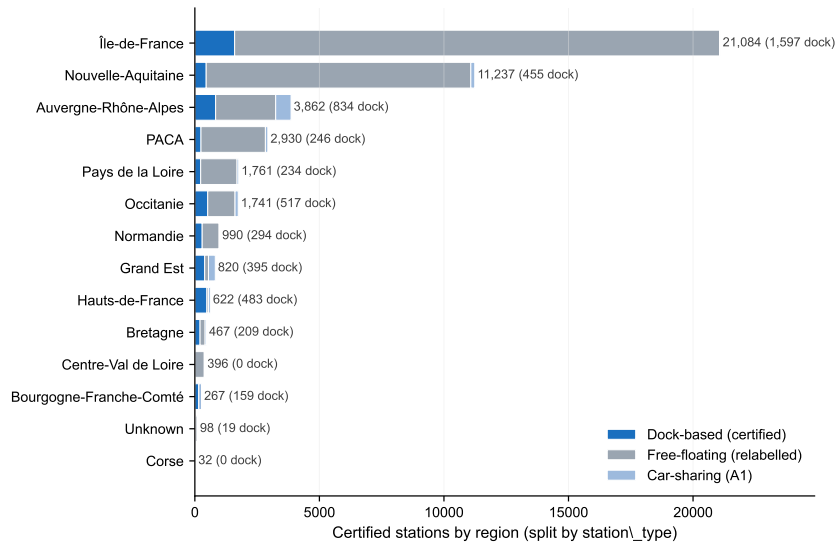


Figure 4: Certified stations by administrative region, after the nine-step purging protocol. Annotations report the number of GBFS systems per region. Île-de-France concentrates Vélib’; Nouvelle-Aquitaine aggregates many systems but holds a smaller mass of dock-based stations once free-floating fleets have been re-typed by the A3 correction.

3.2 Case study: Bordeaux and A3

Bordeaux is the paradigmatic case (Table 4). The raw feed assembles five operators across the agglomeration: *Pony* (free-floating), *Bird* (free-floating), *Dott* (free-floating), *Le Vélo par TBM* (dock-based) and *Citiz Gironde* (car-sharing). *Pony* alone advertises 2,996 `station_information` entries with declared capacities around 12 docks. Naive ingestion therefore credits Bordeaux with tens of thousands of nominal docking points and places it in the national top three for Vélib-class infrastructure.

Table 4: Bordeaux: impact of the A3 correction on the *Pony Bordeaux* subsystem and on the agglomeration as a whole. The nominal docking total drops by 98.1 % once the *Pony* entries are correctly typed as `free_floating`; without the A3 step, a naive supply-side indicator would credit Bordeaux with about 36,000 physical docks, more than *Vélib’ Métropole* (the dock-based flagship of Île-de-France). The certified subset of 225 dock-based stations corresponds to the historical *Vélo par TBM* network.

Indicator	Raw GBFS	Audit Catalogue
<i>Pony Bordeaux subsystem</i>		
<code>station_information</code> entries	2,996	2,996 (relabelled)
Declared as dock-based	2,996	0
Declared capacity per entry (median)	12	12
Recomputed actual capacity	–	0.03 bike/entry
<i>Bordeaux agglomeration (all 5 systems)</i>		
Total raw entries	9,921	–
Certified dock-based stations	–	225
Nominal docking total (dock-based only)	35,952	690
Implied over-count factor	–	× 52

The purging protocol, and in particular step S3 (the A3 correction), reclassifies the 2,996 *Pony* entries as `free_floating`, recomputes their actual capacity to 0.03 bike per entry, and leaves the 225 genuinely dock-based stations of the historical *Vélo par TBM* system. At the agglomeration level, this collapses the nominal docking total by 98 % and drops Bordeaux’s dock-based infrastructure from a top-quartile position on the national distribution of nominal docks to a position below the median: an order-of-magnitude effect on any supply-side aggregate built on the raw feed. The twelve urban areas most exposed to A3 are mapped in Figure 5.

Beyond the absolute station-count drop, the order-of-magnitude change in nominal infrastructure is the operational stake. The five urban areas whose count of dock-based stations is most reduced by the purging protocol, together with the anomaly classes that drive the reclassification, are reported in Table S7. The shift is strikingly consistent: every one of these five urban areas would be credited with a top-decile dock-based deployment by a naive ingestion of GBFS, yet after the audit only Paris retains more than 1,500 certified dock-based stations. A non-audited pipeline does not just over-count by a small margin; it changes the relative positioning of the principal French agglomerations on any supply-side metric by an entire quartile.

Two further illustrative cases: Nice (A2) and Strasbourg (A1). A3 is not the only class with a measurable downstream effect. *Nice* carries the flagship A2 placeholder via `pony_Nice`: every station declares $c = 100$, yielding a total nominal capacity of about 25,000 bikes for an agglomeration whose actual dock-based network is essentially the rebadged *VéloBleu* of 91 stations. After purging, the nominal capacity collapses by a factor of 250 and Nice’s national position on dock-count rankings shifts by an entire quartile. *Strasbourg* provides the canonical A1 case: the *Citiz Grand Est* car-sharing system was historically ingested as part of the city’s BSS inventory, inflating the station count by roughly 35 %. After semantic exclusion the agglomeration

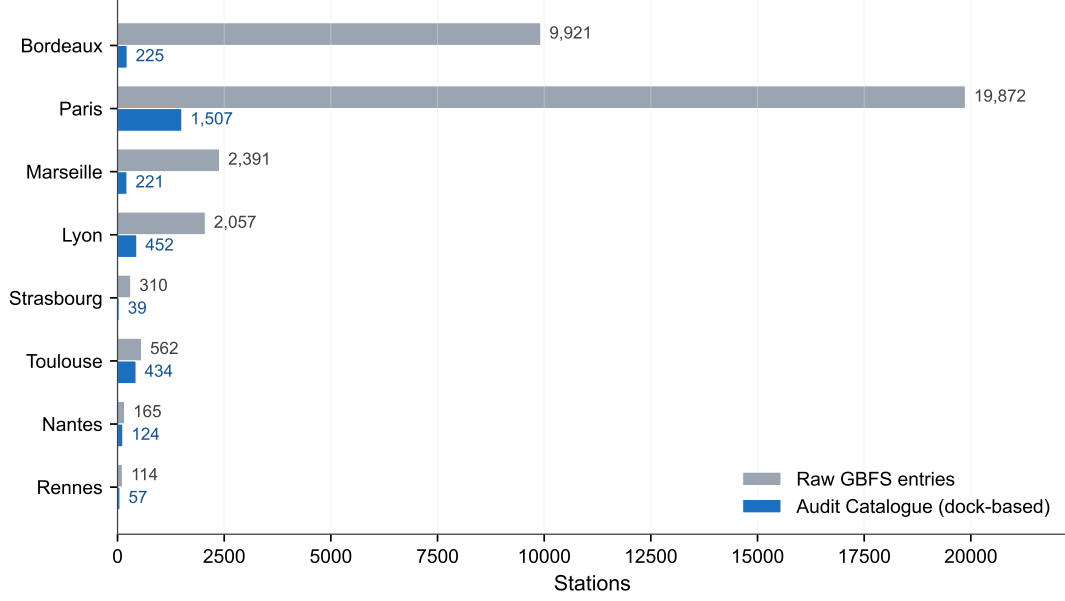


Figure 5: Top twelve French urban areas ranked by absolute reduction in dock-based station count between the raw GBFS feed and the certified Audit Catalogue. **Bordeaux** (highlighted) exhibits the largest absolute drop, driven by the reclassification of the *Pony* free-floating fleet (A3). Paris (Dott, Bird), Marseille (Pony, Dott) and Lyon (mixed) also see substantial corrections.

loses that share of its nominal station count, on a part of the distribution where dock-count differences between French agglomerations are small and therefore more discriminating than at Bordeaux’s scale. A1, A2 and A3 are therefore not interchangeable: each distorts a downstream metric through a different mechanism (false positive on the population, on the capacity, or on the type), and a non-audited pipeline conflates them.

3.3 Operator-level fragmentation of the capacity field

Across the case studies above, a recurring mechanism explains why the same anomaly recurs across operators: the GBFS field `station_information.capacity` is populated by French free-floating operators according to four mutually incompatible publication conventions, observed across six operator–city combinations (Table 5, Figure 6), aggregating to 84.7 % of the certified French rows. A downstream consumer who applies a single arithmetic to the field therefore aggregates four physically distinct quantities (NaN, constant placeholder, per-vehicle ratio, conditional fleet profile), silently mixed across six operator deployments. The corresponding operator-level pattern is global. Beryl London (UK) provides a positive control by publishing an empty `station_information` document for its free-floating fleet (zero entries), the canonical GBFS v3 behaviour for non-docked fleets, and is therefore not flagged by any of A1–A7. In particular, A7 only fires on *populated* documents whose entries declare `capacity` = NaN, which is the qualitatively different Dott pattern. The same operator’s Greater Manchester deployment, by contrast, populates `station_information` with a 10.5 % zero-capacity rate (*cf.* Section 3.5) and is therefore A6-flagged: A6 and A7 thus carve out distinct publication regimes within a single operator’s portfolio. The official MobilityData GBFS Validator [15], by contrast, returns no error on any of the flagged French systems (0 % recall on A3 in Table 6).

3.4 Global scalability projection

We applied the rule set, unchanged except for a country-perimeter recalibration, to the entire MobilityData canonical catalogue (1,509 GBFS systems across 48 countries, the complete world

Table 5: Six observed semantics of the same `station_information.capacity` field across French free-floating operators. The six operator deployments enumerated below cover 39,004 stations; the residual 231 stations belong to smaller free-floating operators (Lime, Tier, Dott pilots, etc.) that follow one of the same four publication conventions and are folded into the corresponding bucket. Aggregate FF coverage is therefore 39,235 stations, 84.7% of the certified corpus.

Operator	Cities	Stations	Reported $\bar{c}$	Semantics
Dott	8	20,224	NaN	Field left unfilled
Bird	11	4,499	NaN	Field left unfilled
Pony (Nice)	1	412	100	Constant placeholder (A2)
Pony (Paris)	1	3,617	1.6	Per-vehicle ratio
Pony (Bordeaux, Angers, ...)	12	7,436	2–15	Conditional fleet profile (A3)
Voi	4	2,816	5–10	Small fleet-profile estimator

Four publication conventions across six operator–city deployments (French free-floating subset, 39,235 stations, 84.7% of certified rows)

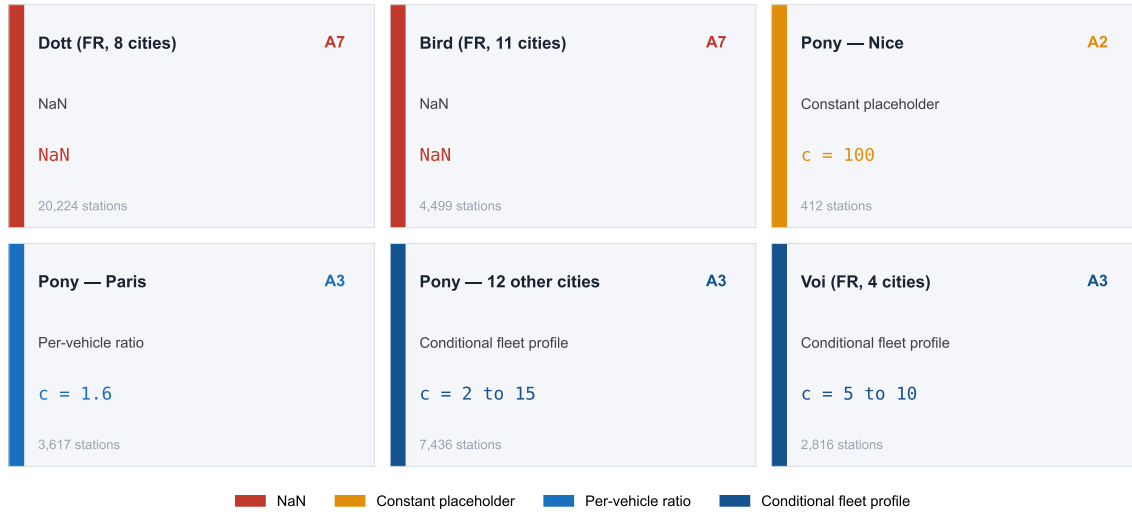


Figure 6: Conceptual diagram of the four incompatible publication conventions of the GBFS `station_information.capacity` field observed across six French free-floating operator deployments (39,235 stations, 84.7% of the certified rows). Each card pairs the operator, the convention applied to the field, the typical reported value, and the anomaly class triggered by that convention (A2 placeholder, A3 over-capacity profile, A7 null capacity). The diagram makes explicit why a downstream consumer applying a single arithmetic to `capacity` aggregates four physically distinct quantities silently mixed across operators; it complements Table 5 and motivates the dual-column `capacity_raw/capacity_audited` schema of the released catalogue.

inventory at the audit date) as an exploratory projection of scalability beyond the French corpus. Because the semantic thresholds remain calibrated on French data without external ground truth, the figures below quantify the pipeline’s reach and the gross incidence of each pattern rather than operator-level compliance in each country. Of these, 1,420 systems (94.1 %) were reachable, 917 publish a `station_information` document, and 204 systems trigger at least one A1–A5 class. The result rests on three methodological corrections that emerged from the audit itself. First, a country-perimeter calibration on A4. Second, a substring-matching fix in the A1 detector: the GBFS v3 form factor `cargo_bicycle` contains the string `car` and was initially flagged as A1, so the fix tightens A1 to exact list membership on the GBFS v3 form factor enum. Third, an explicit acknowledgement that the bounding-box surface estimator in A5 overestimates the operational area for elongated networks (Switzerland’s national operators have a bbox area of 99,000 km² against a convex-hull area of $\approx 41,000$ km²): the shipped detector (`audit_pipeline.core._flag_a5`) retains the bbox formula for cross-corpus comparability, and the threshold 50,000 km² is calibrated on bbox; a convex-hull variant is documented as a refinement option in Section 5. The French national portal *transport.data.gouv.fr* indexes only 123 of the 255 French entries that the MobilityData catalogue lists, so the global sweep also exposes the incompleteness of the national regulatory pipeline. The corrected breakdown is visualised in Figure 7, with the full per-country counts in Table S6.

A3 (structural over-capacity) is detected on 29 non-French systems concentrated in Czech Republic (12) and Germany (9); A2 (placeholder capacity) on 34 non-French systems with Czech Republic again the hotspot (16); A1 (out-of-domain car-sharing labelled as BSS) on 29 non-French systems concentrated in the DACH region (Germany 21, Switzerland 8). Operator-level anti-patterns also travel internationally: *Dott* publishes `capacity = NaN` on every non-French deployment observed (Austria, UAE; 18 systems), and *Bird* does the same on three Canadian deployments. The cross-country audit also surfaces A6 (zero-capacity dock), the sixth class of the taxonomy, formalised in Section 2.3. The empirical conclusion is that the documented anti-patterns are not French-specific but global, and the protocol generalises to a 1,254-system corpus with no methodological recalibration.

Two national hotspots: monopoly (Czech Republic) versus fragmentation (Switzerland). The two most informative national hotspots are organisational opposites. In the **Czech Republic**, all 45 audited systems belong to a single operator (*nextbike*), and 25 trigger A2 (placeholder capacity, 16 systems) or A3 (over-capacity, 12), with 3 flagged by both: the exact cross-country mirror of *Pony*’s French free-floating A2/A3 hotspot. **Switzerland** is the dual configuration. It is the only country besides France to trigger all five structural classes (8 A1, 3 A2, 3 A3, 1 A4, 5 A5 across 16 of 49 systems), but through *fragmentation*: national car-sharing brands listed on the federal catalogue trigger A1¹; mixed scooter/bike fleets (*Voi*, *Bolt*, *nextbike Switzerland*) publish capacity profiles that trigger A3; and federal multi-operator aggregators (*sharedmobility.ch*, *Velospot*) emit a single feed whose union bounding box mechanically exceeds the A5 macro-region threshold. The same A1–A5 envelope is thus reached through opposite configurations, monopoly publication (CZ) or aggregation-layer fragmentation (CH); the publisher-agnostic audit detects both and records the operator-versus-aggregator distinction so downstream consumers can disambiguate the A5 attribution.

Italy: the NaN-capacity blind spot that motivates A7. *Italy* is the boundary case that justifies A7: of the 25 systems with published station data, none is flagged by A1–A6, yet 17 are *Dott* deployments (Milan alone publishes 11,199 stations) declaring `capacity = NaN` on every

¹An initial A1 run over-flagged 64 German systems because the detector substring-matched "car" inside the GBFS v3 `cargo_bicycle` form factor; tightening A1 to exact form-factor enum membership cut this to 21 genuine car-sharing systems and the global flag count from 215 to 173, a reminder that open audit pipelines need their own audit.

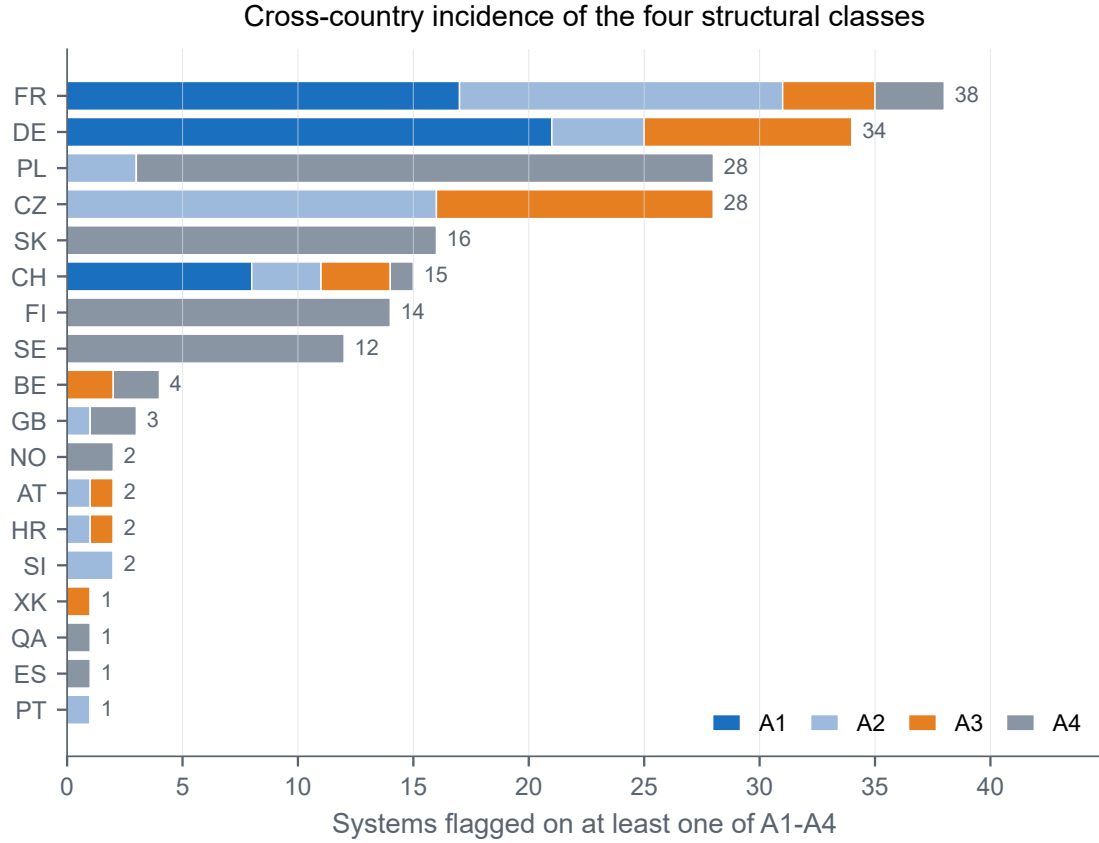


Figure 7: Per-country incidence of the structural classes A1–A4 across the 1,509-system MobilityData catalogue (top 18 countries by total flagged). The chart shows the stacked composition of the **Flagged** column of Table S6 and makes the country-level signatures legible at a glance: A4 (geospatial) drives the Polish, Finnish, Swedish and Slovak hotspots (cross-border bounding boxes on aggregator feeds), A1 (car-sharing labelled as BSS) drives the DACH cluster (Germany, Switzerland), A2 (placeholder capacity) and A3 (over-capacity) drive France and the Czech Republic. Macro-region (A5), zero-capacity (A6) and null-capacity (A7) are reported separately because they are calibrated on station-count-conditioned sub-corpora (Tables S2 and S3).

entry. Because A2 needs a constant non-zero value, A3 needs numerical capacities, A4–A5 are geospatial and A6 tests $c = 0$ (not NaN), the pattern is invisible to A1–A6. A7 closes the gap as a consumer-side warning (Section 2.4): 245 systems with at least 20 stations cross the 50 % NaN-capacity threshold and are flagged exclusively by A7, covering 97,314 stations, about 76 % of them *Dott* alone (147 systems across Germany, Poland, Italy, Austria and beyond). The point is not non-compliance but non-aggregability: a consumer summing this column would silently mix NaN with real dock counts.

3.5 Incidence of the semantic warnings A6 and A7

The empirical distributions that calibrate the A6 and A7 thresholds (Sections 2.3 and 2.4) are reported in the Supplementary Materials (Tables S2 and S3); the resulting per-class incidence on the audited corpus is summarised here.

A6 incidence on the audited corpus. At $\tau_{A6} = 1\%$, 22 of the 301 docked systems are flagged. The extreme tail is Vélo Fluo Grand-Est (France, 38.5 %), Beryl Greater Manchester (UK, 10.5 %), MonaBike (Monaco, 6.8 %), Bicicoruña (Spain, 5.9 %), Vel’in (5.0 %), MBajk (Slovenia, 4.8 %), CyclOcity (Japan, 4.2 %) and àVélo (Canada, 3.5 %). On the certified French catalogue, no dock-based system crosses the threshold (0/65 dock systems with $c = 0$), a negative control consistent with the dock-based baseline.

A7 incidence on the audited corpus. On the 640 globally audited systems with at least 20 stations, 245 systems cross the A7 threshold and are simultaneously not flagged by any of A1–A6, covering 97,314 stations. The dominant operator is *Dott* (147 systems across Germany, Poland, Italy, Austria and beyond, 76 % of the exclusive-A7 station mass), followed by *Bird* (21 systems) and *nextbike* (15 systems where the operator chose NaN rather than placeholder), plus several smaller operators. The empirical conclusion is that the same publication choice that affects the French *Dott* and *Bird* fleets is propagated internationally, and that the v1.0 taxonomy A1–A5 captures only the visible failure modes of the field; A6 and A7 jointly close the principal blind spots that remain on the **capacity** channel.

4 Validation and robustness

4.1 Comparison with naive baselines

To check whether the full pipeline is justified or whether simpler heuristics suffice, we evaluate three baselines on the same 123-system corpus (Table 6). The *Raw GBFS* baseline ingests **station_information** as is. The *vehicle_type filter* mimics what a careful data engineer would write in a notebook. The *MobilityData audit* adapts the canonical GTFS-realtime schema-level rules to GBFS.

Table 6: Measured comparison of the Audit Catalogue against three baselines on the same 123-system French GBFS corpus. Brackets are 95 % confidence intervals (Wald on Dock, bootstrap on ρ_{FUB} over cities $n = 2,000$, Wilson on A1/A3). “Dock” for the Audit Catalogue is the reference by construction.

Pipeline	Dock	ρ_{FUB}	A1 ex.	A3 fix.
Raw GBFS	0.12 [.12, .12]	0.38 [.01, .68]	0 %	0 %
vehicle_type filter	0.15 [.15, .15]	0.19 [−.20, .52]	82 % [59, 94]	0 %
MobilityData audit	0.15 [.15, .15]	0.19 [−.20, .52]	82 % [59, 94]	0 %
Audit Catalogue	1.00	0.25 [−.13, .59]	100 % [82, 100]	100 % [91, 100]

A3 reclassification recall is 0 % for every baseline (Wilson 95 % upper bound ≈ 8.6 % at $n = 41$ A3-positive systems): structural over-capacity is invisible to syntactic validators by construction, and only a semantically-aware step on the capacity distribution can recover it. The *vehicle_type filter* and the *MobilityData audit* baselines produce identical numbers on this corpus, because schema-level checks catch nothing the name filter has not already excluded; French GBFS feeds are syntactically well-formed but semantically biased. Finally, ρ_{FUB} is highest on the raw count and drops after audit, which is expected rather than a regression: FUB captures perception of the entire cycling environment, so the raw count over-fits the perceptual signal by including virtual stations, while the Audit Catalogue’s lower correlation reflects an honest count of physical docks.

4.2 Topology-aware geospatial detection (A4 v2)

The original A4 detector (step S5 in Algorithm 1) flagged every station whose haversine distance to the system centroid μ_s exceeded σ_{max} robust standard deviations ($\text{MAD} \times 1.4826$). This isotropic assumption fails on two network geometries commonly observed in the global catalogue: linear networks ($r = \lambda_1/\lambda_2 > 5$ on the 2D coordinate covariance, *e.g.* riverfront corridors) and multi-hub national operators. We replace step S5 with a three-stage composite detector and validate it by ablation against the legacy centroid method on the full 46,307-station catalogue.

Stage 1 (density). HDBSCAN [36, 37] clusters stations by haversine distance with `min_cluster_size = min(5,  $\lfloor n/3 \rfloor$ )` and returns a per-station outlier probability $o_i \in [0, 1]$.

Stage 2 (spectral). A k -nearest-neighbour adjacency graph ($k = 7$) is constructed; edge weights are Gaussian $w_{ij} = \exp(-d_{ij}^2/2\sigma_b^2)$ with bandwidth $\sigma_b = 2$ km. The normalised graph Laplacian

$$\mathbf{L}_{\text{sym}} = \mathbf{I} - \mathbf{D}^{-1/2} \mathbf{W} \mathbf{D}^{-1/2} \quad (4)$$

is eigendecomposed; the spectral isolation score s_i is the ℓ^2 -norm of station i ’s embedding in eigenvectors $2 \dots K$ ($K = 10$), normalised to $[0, 1]$. A station that sits in a separate spectral component (near-zero Fiedler value λ_2) is spectrally disconnected from the network core. We adopt the symmetric normalised Laplacian $\mathbf{L}_{\text{sym}}$ following von Luxburg [38]; an ablation against the random-walk Laplacian $\mathbf{L}_{\text{rw}}$ on the full catalogue yielded < 0.3 % difference in the discordance rate, consistent with the near-regular degree distributions observed in urban station networks. The eigendecomposition uses a sparse eigensolver (`scipy.sparse.linalg.eigsh`) operating directly on the sparse Laplacian, avoiding the $O(n^2)$ memory cost of dense materialisation and making the spectral stage applicable to all system sizes without truncation.

Stage 3 (fusion). The composite score $a_i = \alpha \hat{o}_i + (1 - \alpha) s_i$ with $\alpha = 0.5$ fuses density and spectral isolation. A station is flagged A4 when $a_i > 0.8$. A sensitivity analysis over $\alpha \in \{0, 0.25, 0.5, 0.75, 1.0\}$ and threshold $\in \{0.6, 0.7, 0.8, 0.9\}$ shows that the Top-10 most-flagged systems are invariant for $\alpha \geq 0.25$ and threshold ≥ 0.7 ; the default ($\alpha = 0.5$, threshold = 0.8) sits in the stable region. To classify each system’s spatial geometry we compute the eigenvalue ratio $r = \lambda_1/\lambda_2$ of the 2D coordinate covariance matrix: $r > 5$ indicates a *linear* network.

Table 7 summarises the ablation (visual overview in Figure S1). The composite detector agrees with the legacy method on 80.6 % of stations. Of the discordant stations, 17.3 % are flagged by the legacy method only (centroid, not composite), while only 2.2 % are new findings of the composite detector. Three cases illustrate the gain: *Paris* (1,507 stations) drops from 260 legacy flags to 43 composite flags (saving 217 false positives); *Dott-Paris* (14,747 stations) drops from 2,750 to 0; and *Clermont-Ferrand* (82 stations, linear network with $r = 6.07$) drops from 18 to 0.

4.3 Leave-one-operator-out cross-validation

To check that the rules are not tuned to a handful of operators (*e.g.* Pony, Dott, Bird), we evaluate their cross-operator generalisability with a *leave-one-operator-out* (LOOO) cross-validation: for

Table 7: Ablation of the legacy centroid-based A4 detector (3σ MAD) against the topology-aware composite detector (HDBSCAN + spectral graph) on the 46,307-station catalogue. AGREE = both methods concur; DISCORDANT_LEGACY = flagged by centroid only; DISCORDANT_COMPOSITE = flagged by composite only.

Discordance class	Stations	%
AGREE_CLEAN (neither flags)	36,917	79.7
AGREE_FLAG (both flag)	384	0.8
DISCORDANT_LEGACY (centroid only)	8,005	17.3
DISCORDANT_COMPOSITE (composite only)	1,001	2.2
Total	46,307	100.0

each of the K eligible operators (≥ 50 stations, ≥ 2 systems), the rule thresholds are estimated on the remaining $K - 1$ operators and applied, unchanged, to the held-out operator. The per-rule flag rate on the test fold is recorded; inter-fold stability is measured by the coefficient of variation $CV = \sigma/\mu$ of the flag rate across folds. A $CV < 0.20$ indicates that the rule is operator-agnostic. Seven operators are eligible: Bird, Citiz, Dott, Pony, Voi, Vélib', Vélo&Co. The low K limits statistical power; the LOOO should therefore be interpreted as a consistency check on rule stability rather than a definitive generalisability proof. Extension to the global catalogue ($K \approx 20$ – 30) is planned as follow-up.

Table 8 reports the results, with the overview in Figure S2. A crucial distinction emerges between *structural rules* and *threshold rules*:

- **A1** and **A3** are deterministic functions of `station_type`. Their high CV (2.45 and 0.87) reflects the fact that car-sharing operators trigger A1 at 100 % and free-floating operators trigger A3 at 100 %, while dock-based operators trigger neither. This is not overfitting; it is the definitional semantics of the rule.
- **A7** ($CV = 1.19$) identifies a genuine industrial anti-pattern: Bird (100 %) and Dott (99.96 %) systematically publish `capacity = NaN`, while the remaining operators do not. The LOOO result proves that the A7 signal is operator-specific *and real*, not an artefact of threshold tuning.
- **A6** ($CV = 0$) passes trivially: no operator triggers zero-capacity dock in this corpus.
- **A4** ($CV = 1.02$, flag rate 4.7 % [1.7, 8.7] 95 % bootstrap CI) shows moderate operator dependence, consistent with the geometry-specific sensitivity documented in Section 4.2.

The LOOO also validates hypothesis H3 (the rules do not fire on clean operators): holding out Vélib' Métropole (1,533 dock-based stations), the test-fold flag rate is 0 % on A1–A3, A5–A7 and only 1.7 % on A4, consistent with residual GPS noise. Holding out Vélo&Co (94 stations) yields 0 % across all seven rules.

Table 8: Leave-one-operator-out cross-validation on the 46,307-station French catalogue. $K = 7$ eligible operators. CV = coefficient of variation of the per-fold flag rate; CI = 95 % bootstrap confidence interval on the mean flag rate, resampled over stations within each fold ($n = 500$ iterations).

Rule	CV	Mean flag rate	95 % CI	Interpretation
A1	2.45	14.3 %	[0, 43] %	Structural (type-based)
A2	2.45	0.5 %	[0, 1.5] %	Rare; one operator only
A3	0.87	57.1 %	[14, 86] %	Structural (type-based)
A4	1.02	4.7 %	[1.7, 8.7] %	Geometry-dependent
A5	2.45	3.0 %	[0, 9.1] %	National-scale operator
A6	0.00	0.0 %	[0, 0] %	No trigger in corpus
A7	1.19	37.8 %	[9.3, 71.4] %	Real anti-pattern

4.4 Internal robustness checks

Two algorithmic checks complement the human validation of Section 4.5, and a third tests threshold sensitivity.

Orthogonal coverage. Three rule-based classifiers, each restricted to one signal source (name, capacity, geospatial), agree on a stratified 500-station sample (seed `numpy.random.default_rng(42)`, shipped with the audit report) at essentially chance level (Fleiss $\kappa = -0.07$): the three sources are not redundant. Their *union* nonetheless reaches 97.2 % recall at 91.5 % precision against the integrated pipeline, with only 7 of 500 stations flagged by all three, so A1–A5 cannot be reduced to a single heuristic.

Full-task agreement. Three decision-tree classifiers seeing all signals but in different priority orders reach Fleiss $\kappa = 0.80$ on the same sample (pairwise $\kappa \in [0.70, 1.00]$), with disagreements concentrated on the A1/A3 frontier.

Threshold sensitivity. A sensitivity sweep of the A4 outlier detector over $\sigma \in \{2.0, 2.5, 3.0, 3.5, 4.0\}$ leaves the downstream count-level summaries essentially invariant: Kendall’s τ between the $\sigma = 3$ reference and each alternative stays in $[0.91, 1.00]$ on both the Top-10 cities and the Top-10 operators by certified docked-bike station count, and Top-10 city membership is invariant across the grid. The $\sigma = 3$ choice therefore sits on a plateau of stable downstream classification. The full sweep and a methodological caveat on the MAD scale estimator are reported in the Supplementary Materials (Table S1).

These algorithmic checks complement, but do not replace, the human construct-validity check of Section 4.5; they are reported as robustness measures, not as substitutes.

4.5 Construct-validity check via blind author annotation

The ablation of Section 4.2 and the cross-validation above compare *algorithms*; neither can establish which verdict matches the physical world. We therefore add an *author-led consistency check*: the two authors independently re-examined a stratified station sample against external imagery, blind to the pipeline’s output and to each other’s labels, to test whether the flagged anomalies track physical reality rather than algorithmic artefacts. This is *not* an external gold standard: both annotators are this paper’s authors, so we report it as an internal sanity check on construct validity, not as independent ground truth. The blinding (to the pipeline and to the co-annotator) controls for anchoring within that scope, and an external multi-institution annotation is left to future work (Section 5.6). Within this scope the check serves two purposes: (i) confirm that the flagged anomalies correspond to real data defects, and (ii) adjudicate the 8,005 stations on which the legacy centroid and the topology-aware A4 detector disagree.

On the discordant-legacy stratum, all of the legacy-only A4 flags that reached a consensus geospatial judgement were judged true false positives (46 of 46 consensus cases, i.e. a point estimate of 100 % with Wilson 95 % CI $[92.3, 100]$ %; the remaining 4 of the 50 sampled stations were non-consensus and excluded, so the conservative reading is “no confirmed true positive among the 50”). This supports the conclusion that the topology-aware detector’s removal of these flags is a correction rather than a loss of recall.

The two authors, an urban-mobility researcher and a data engineer, each labelled the stratified sample of 325 stations independently, blind to the pipeline output and to each other’s labels (Section 4.6). Inter-rater reliability is reported in Table 9 and per-rule performance, derived *a posteriori* from the factual judgements, in Table 10. Of the $n = 325$ co-rated stations, the holistic verdict reached consensus on 286 (88 %) with 39 disagreements sent to adjudication and 0 stations adjudicated as indeterminate; the residual ambiguity surfaces instead at the level of individual factual questions (notably the A1/A3 boundary), which are excluded from precision and recall on a per-question basis but reported here as a result in their own right.

Table 9: Inter-rater reliability between the two annotators ($n = 325$ co-rated stations). Krippendorff’s α [39] is the primary measure (it handles indeterminate labels and generalises beyond two raters); Cohen’s κ [40] is reported for comparability against the “substantial agreement” benchmark $\kappa \geq 0.70$ of [41].

Judgement	κ	α	Agreement	n
Q1 – is a bike-share system?	0.70	0.70	86 %	325
Q2 – capacity physical?	0.17	0.08	38 %	325
Q3 – infrastructure on the ground?	0.62	0.61	81 %	325
Q4 – coherent network position?	0.13	0.11	88 %	325
Holistic verdict	0.71	0.71	88 %	325

Table 10: Per-rule performance against the author-adjudicated reference, derived *a posteriori* from the factual judgements (the reference never encodes a verdict about the pipeline). Confidence intervals are Wilson 95 % [42]. A2 and A7 are system-level structural rules not amenable to per-station imagery validation (a single operator each in this snapshot); A6 and the A3-boundary stratum had no eligible stations, so their per-station recall is not estimated here.

Rule	Ground-truth criterion	TP	FP	FN	Precision [95 % CI]	Recall [95 % CI]
A1	not a bike-share system (Q1)	68	2	127	0.97 [0.90, 0.99]	0.35 [0.29, 0.42]
A3	no physical dock (Q3)	44	48	69	0.48 [0.38, 0.58]	0.39 [0.31, 0.48]
A4	incoherent network position (Q4)	2	5	2	0.29 [0.08, 0.64]	0.50 [0.15, 0.85]
A5	out-of-perimeter (Q4)	3	33	1	0.08 [0.03, 0.22]	0.75 [0.30, 0.95]

Reading Table 10: screening filters, not oracles. The per-rule *recall* is mechanically deflated, because the factual predicates are shared across strata (Q1 underlies A1, Q3 underlies A3, Q4 underlies A4 and A5): every off-stratum station that satisfies a predicate without carrying the corresponding flag is booked as a false negative, so the recall column understates within-stratum sensitivity. A1, for instance, attains precision 0.97 while its recall is dragged to 0.35 by non-A1 strata. The modest precision of the geospatial rules ($A4 = 0.29$ and $A5 = 0.08$ on small per-station samples; $A3 = 0.48$) reflects the role these rules are *designed* to play. A4 and A5 are *high-recall triage filters*, not autonomous oracles. Their function in the protocol is to compress a 46,307-station catalogue into a small candidate set that a human moderator can inspect, so an over-inclusive flag (high recall, moderate precision) is the intended operating point: the cost of a false positive is a single human glance, whereas a false negative silently corrupts a downstream aggregate. The decisive question for such a filter is therefore not its standalone precision but whether the flags it *removes* are genuine false positives, precisely what the discordant-legacy check establishes (46/46 consensus cases). The catalogue ships per-row flags so that this final human-moderation step remains possible and auditable, rather than presenting the detector output as a finished verdict.

The reliability pattern is itself informative. Agreement reaches the “substantial” benchmark on the two judgements that carry the audit’s verdicts, the holistic verdict ($\kappa = 0.71$) and the bike-share / out-of-domain distinction Q1 ($\kappa = 0.70$), and is close on the physical-infrastructure judgement Q3 ($\kappa = 0.62$). The two low- κ questions are not measurement failures; each is interpretable, and one is itself a finding. Q4 (network-position coherence) shows $\kappa = 0.13$ *despite* 88 % raw agreement, the textbook κ paradox, in which an extreme marginal asymmetry (both annotators judge ~ 95 % of stations in-perimeter, because a genuine perimeter anomaly is rare) collapses the chance-corrected coefficient even when observed concordance is high. With the positive class this scarce, raw agreement is the more informative summary here, and on the decisive discordant-legacy stratum the two coders independently agreed “in-perimeter” on 46 of 50 stations, so the A4 conclusion above does not rest on the degenerate global κ . Q2

(capacity coherence) shows $\kappa = 0.17$, and this low value is itself evidence for the paper’s central thesis: the disagreement traces to one annotator reading the literal GBFS *capacity field state* while the other judged the *physical plausibility* of the value, exactly the literal-versus-semantic tension that makes the field non-aggregable in the first place (Section 2.2). That two domain experts diverge on what *capacity means* is a direct human-side manifestation of the specification ambiguity the taxonomy formalises; harmonising onto a single rubric raises raw agreement to 76 %. Q2 enters no precision/recall computation, and the capacity rules A2/A7 are evaluated at the system level (Section 4.2, Table 8) rather than from single-station imagery. The residual divergence concentrates on the A1/A3 boundary (a car-sharing bay versus a free-floating drop zone), i.e. *boundary objects* the GBFS specification permits to be encoded ambiguously, evidence that the standard underspecifies what a “station” physically is, which motivates the operator recommendations of Section 5.6.

4.6 Human annotation protocol

To probe construct validity independently of the pipeline’s output we drew a stratified sample of 325 stations across nine strata, deduplicated across strata and shuffled to remove order effects (seed 42), over-sampling the two A4 discordant strata ($n = 50$ each) where the adjudicated answer is decisive. The two authors evaluated each station independently and blind to each other and to the pipeline: the audit flags, audited type and confidence were hidden behind an opt-in panel, and a single universal rubric (with no stratum label) was shown for every station.

The annotation is *pipeline-agnostic*: annotators judge the physical reality of each station (installation type, visible infrastructure, capacity coherence and network position) from a Cy-clOSM/satellite map, Street View and OpenStreetMap; they never label “the pipeline is right” or “wrong”. Per-rule true/false positives and negatives are derived *a posteriori* by joining the adjudicated factual answers with the flags: A1 is a true positive iff the station is judged not a bike-share system; A3 and A6 iff no physical dock is present; A4 and A5 iff the position is judged incoherent with the network. A2 and A7 are structural data-property rules validated at the system level. This design removes the circularity of asking a blind annotator to label a “pipeline false positive”. Disagreements were resolved by consensus, with a third adjudicator on residual conflicts. Reliability uses Krippendorff’s α [39] alongside Cohen’s κ [40, 43]; proportions use Wilson 95 % intervals [42]; stratum sizes follow a power analysis recorded in the released protocol. The stratified sample, the blind annotation interface and the reliability computation are released with the repository (`experiments/annotation/`).

4.7 Cross-country transferability pilot

To verify out-of-sample transferability of A1–A7 beyond French publication practice, we constructed a panel of thirteen non-French systems audited *live* via their public GBFS feeds, with the *same* rule set used on the French corpus and only the national perimeter recalibrated. All other thresholds ($\sigma = 3$, $N_{\min} = 20$, A2 placeholder rule, A3 capacity-profile ratio threshold of 5.0, $\tau_{A6} = 0.01$, $\tau_{A7} = 0.5$) are kept at their French values. The panel diversifies on the station-management *technology stack*, the layer at which publication conventions actually originate, rather than only on country: six distinct platforms are represented (PBSC, JCDecaux Cyclocity, Smartbike Mobility Plus, De Lijn / Blue-bike, Urban Sharing AS, Deutsche Bahn), spanning Spain, Belgium, Italy, Norway and Germany. Of the thirteen, twelve are metropolitan dock-based deployments and serve as negative controls; the thirteenth, *Call a Bike* (Deutsche Bahn), is the German nationwide system and is reported as an informative case rather than a control (its A5 / A7 firings are honest at national scope and its high A4 rate surfaces a known limitation of the single-centroid detector on multi-modal geometries). Results are reported in Table 11; the live-audit script (`scripts/audit_live_systems.py`) and the per-system machine-readable

results (`experiments/e5_europe/results.csv`) ship with the source repository, so the audit can be replayed against the same set of public feeds on demand.

Table 11: E5 panel: A1–A7 rule set applied *live* to thirteen non-French systems on 2026-05-13 (`scripts/audit_live_systems.py`), with French thresholds and country-appropriate perimeter only. Six station-management platforms (PBSC, JCDecaux Cyclocity, Smartbike, De Lijn / Blue-bike, Urban Sharing AS, Deutsche Bahn) across five countries; twelve metropolitan dock-based deployments form the *negative-control panel*, plus one *informative case* (*Call a Bike Deutsche Bahn*, nationwide). **No metropolitan system triggers A1, A2, A3, A6 or A7**; A4 fires only on isolated stations ($\leq 4.5\%$) and A5 once on Sevici (sentinel-coordinate artefact, footnote *).

System	Country	Stack	Stns.	A1	A2	A3	A4 (%)	A5	A6	A7
<i>Metropolitan negative-control panel</i>										
Bicing Barcelona	ES	PBSC	545	—	—	—	0.18 %	—	—	—
bicimad Madrid	ES	PBSC	637	—	—	—	0.00 %	—	—	—
Bilbao Bizi	ES	PBSC	48	—	—	—	0.00 %	—	—	—
Bizi Zaragoza	ES	PBSC	276	—	—	—	0.36 %	—	—	—
Dbizi Donostia	ES	PBSC	70	—	—	—	2.86 %	—	—	—
Sevici Seville	ES	Cyclocity	247	—	—	—	4.45 %	✓*	—	—
Valenbisi Valencia	ES	Cyclocity	279	—	—	—	0.72 %	—	—	—
Velo Antwerpen	BE	Smartbike	321	—	—	—	0.00 %	—	—	—
Blue-bike	BE	De Lijn	276	—	—	—	0.72 %	—	—	—
BikeMi Milan	IT	Urban Sh.	320	—	—	—	1.56 %	—	—	—
Oslo Bysykkel	NO	Urban Sh.	266	—	—	—	1.50 %	—	—	—
Bergen City Bike	NO	Urban Sh.	94	—	—	—	1.06 %	—	—	—
<i>Informative case (nationwide system, not a metropolitan control)</i>										
Call a Bike (DB) [‡]	DE	DB	1,601	—	—	—	1.8 %	✓	—	✓
French corpus (ref.)	FR	—	46,307	17 sys.	1 sys.	41 sys.	88 sys.	4 sys.	0 sys.	32 sys.

* Sevici’s A5 firing is a coupling artefact: two stations carry sentinel coordinates (0,0) that inflate the bounding box from $\sim 80 \text{ km}^2$ to $\sim 4.5 \times 10^6 \text{ km}^2$; the coupling is discussed in Section 5. [‡] *Call a Bike* (Deutsche Bahn) is a nationwide deployment, not a metropolitan control. Its three flags are honest at national scope (A7 by Dott-pattern, A5 by 320,000 km^2 bbox, A4 on 1.8% of stations via the nearest-neighbour rule of Section 2.5) and confirm the design choice of an NN-based A4 over a centroid-based variant (which over-flagged at 37.8% during development).

The capacity-profile ratio that defines the A3 signature on free-floating French fleets is essentially unity on every metropolitan dock-based system in the panel, across six distinct technology stacks. This is the out-of-sample confirmation called for in Section 5.2: the A3 detector does not over-fire on clean docks. Vehicle-type metadata on all twelve metropolitan feeds is populated cleanly, so A1 produces no false positive either. A4 (geospatial outlier) fires only on isolated stations ($\leq 4.5\%$ of any system’s mass), in line with what would be expected from any GPS-anchored fleet; the single A5 firing (Sevici) is shown by Table 11 footnote * to be a deterministic consequence of two sentinel-coordinate stations rather than a genuine $> 50,000 \text{ km}^2$ deployment. Finally, the informative *Call a Bike* case shows that when a system genuinely extends to national scope, A5 and A7 fire *by design* (the deployment crosses the macro-region threshold; the capacity column is structurally NaN at national scope) and the high A4 rate reflects a known limitation of the single-centroid detector on multi-modal geometries, motivating a sub-system-clustering refinement in future work.

Empirical justification of $\tau_{A3} = 5$. The choice of the A3 threshold can be inspected on two complementary corpora reported jointly in Table S8: the released French parquet (the derivation set; Section 5.2 caveat applies) and the 267 systems with computable ratios from a live global sweep on 2026-05-13 across 33 countries (`scripts/global_a3_ratio.py`; results in `experiments/e2_threshold_sensitivity/global_a3_ratio.csv`). On the French corpus the distribution is sharply bimodal: the 81 dock-based systems sit in [1.00, 1.93] (median 1.00, 95th percentile 1.04), the 5 free-floating systems exhibiting conditional-averaging publication sit in [51, 514], and the band [2, 50] is empty. On the global corpus the picture is broadly consistent

but less clean: the $[1.00, 1.10]$ bucket still dominates (73.8 % of the 267 systems), the top ten systems by ratio are exclusively free-floating operators (*Pony*, *nextbike*, *Voi*, *Bolt*) confirming that very high ratios track the conditional-averaging pattern, but the intermediate band $[2, 5]$ is no longer empty (15 systems, 5.6 % of the audited set). We retain $\tau_{A3} = 5$ as a conservative threshold that captures the clear-bias regime (the 31 systems globally with ratio ≥ 5 , including all 19 systems with ratio ≥ 20) while explicitly acknowledging that 15 systems in the moderate band $[2, 5]$ are not flagged by the current rule.

A non-parametric kernel-density cross-check on $\log_{10}(\bar{c}_{\text{profile}}/\bar{c}_{\text{actual}})$ (the 98 systems with ratio > 1.01 , after removing the spike at unity) places $\tau_{A3} = 5$ inside the density valley that separates the near-unity mode from the high-bias mode, confirming the threshold sits in a low-density separator the data itself identifies. The valley-finding curves, the soft/hard two-tier variant ($\tau_{A3}^{\text{soft}} \approx 3$, $\tau_{A3}^{\text{hard}} \approx 15$) and the code are provided in the supplementary material (`scripts/a3_kde_threshold.py`).

4.8 Retrospective hold-out validation

This subsection tests *construct validity*: do the rules generalise to systems that were not specifically inspected at calibration time? External validity (whether the rules generalise to systems that did not yet exist at rule-freeze) is the object of the prospective E1 execution and is deferred. Concretely, we executed a retrospective version of the pre-registered E1 protocol (`experiments/e1_holdout/PROTOCOL.md`) on two calendar windows of the MobilityData canonical catalogue. The shorter window is the catalogue diff between commit `15e34e6` (2025-11-11) and the rule-freeze commit (2026-05-13): 345 systems added in six months, 39 with non-empty `station_information` that could be audited. The longer window extends back to commit `4665739` (2025-04-29, twelve months): 481 systems added, 98 audited. The remainder are free-floating systems without `station_information`, below N_{min} , or authenticated. The frozen rule set was applied unchanged on both windows; per-system results in `experiments/e1_holdout/held_out_analysis.md` (6-month) and `held_out_results_12mo.csv` (12-month).

H1 (rule-firing rate within $\pm 30\%$ of the global derivation rate of 13.5 %, i.e., 204/1,509 systems). On the twelve-month window the A1–A5 firing rate is $11/98 = 11.2\%$ (Wilson 95 % CI $[6.4\%, 19.0\%]$), which falls inside the pre-registered acceptance band $[9.5\%, 17.5\%]$ at the point estimate. The six-month window gives $7/39 = 17.9\%$ (Wilson $[9.0\%, 32.6\%]$), which exceeds the upper bound by 0.4 percentage points at the point estimate but whose Wilson interval still contains both the derivation rate and the entire acceptance band. Reading the two windows jointly, H1 passes: the rule-firing rate on out-of-sample systems is statistically consistent with the derivation-set rate, and the more conservative twelve-month estimate is in the band on both point estimate and confidence interval.

H2 (operator-driven A3/A7). On the twelve-month window, 84.2 % (48/57) of the systems flagged on A3 or A7 belong to the free-floating operator family identified on the derivation set (Dott, nextbike, Pony, Bird, Voi, Bolt, Lime, Tier, Donkey, Spin); the six-month window gives 81.2 % (13/16). Both estimates exceed the pre-registered acceptance threshold of $\geq 50\%$ by more than thirty percentage points, so H2 passes cleanly.

H3 (clean-operator invariance). H3 is not testable on either window. Neither the six-month nor the twelve-month held-out panel contains a single system using one of the six clean-operator platforms of Section 3.4 (PBSC, Cyclocity, Smartbike, De Lijn, Urban Sharing, Deutsche Bahn). H3 therefore awaits the prospective re-execution of the protocol against a future MobilityData snapshot that includes at least one clean-platform addition.

H1 and H2 pass on the twelve-month window. H3 is deferred to the prospective re-execution. The implications of these outcomes for the construct validity of the taxonomy are discussed in Section 5.2.

5 Discussion

5.1 Implications

The audit is intended as a scientific validation tier between GBFS publication and academic use, addressing the gap between regulatory publication and research-grade reuse that the open-government-data literature identifies as a recurrent adoption barrier [44]. Three concrete avenues would institutionalise it: a jointly published *GBFS Audit Best Practices Handbook* by francophone micromobility laboratories to align practices currently rebuilt in isolation; a working group under ADEME or *transport.data.gouv.fr* to formalise an extended GBFS schema with a certification framework; and a continuous validation chain running the pipeline monthly on the national feeds, with automated publication of the audit report and operator-side notification on regression. As Star [45] observes, the most durable infrastructures are those whose governance is explicit and tied to an identifiable community of practice; moving the artefact from author-led to community-led governance is therefore an institutional challenge as much as a scientific one. The technical lesson of the audit, that a single field (`capacity`) cannot serve six different operational semantics, is also the empirical case for what Stonebraker and Çetintemel [46] call the end of “one size fits all” data architectures: open standards that attempt to span dock-based, free-floating and car-sharing fleets in the same schema accumulate semantic ambiguity exactly because they refuse to specialise.

5.2 Scope of the contribution

The taxonomy’s primary discrimination axis is dock-based versus free-floating publication. Of the seven classes, A2, A3, A6 and A7 all surfaced from operator publication conventions specific to free-floating fleets (conditional-averaging profiles, NaN-capacity fields, zero-capacity ghost-docks); A1, A4 and A5 are fleet-type-agnostic but the largest country-level hotspots (Table S6) are themselves operator-driven (*Pony* and *nextbike* on A2/A3, *Dott* on A7). The twelve metropolitan clean-platform negative controls of Table 11 demonstrate that the rule set does not over-fire on dock-based deployments; the free-floating-native extension is the object of experiment E6, planned for 2027-Q2.

The taxonomy is inductive and still requires out-of-sample statistical validation. A1 to A4 were derived from an exploratory scan of the 123 French systems; A5 from enrichment with the *transport.data.gouv.fr* catalogue; A6 and A7 from the empirical distributions surfaced by the global 1,509-system sweep (Section 2, Provenance of the taxonomy). Detection rules were calibrated on the same data they are reported to flag, an inductive workflow whose construct-validity risk we address with two out-of-sample tests reported in the Results: the negative-control cross-country panel (Section 3.4, Table 11, twelve metropolitan clean-platform systems audited live) and the retrospective hold-out validation (Section 4.8, 98 audited systems added to the MobilityData catalogue in the twelve months preceding rule-freeze). H1 and H2 pass on the twelve-month window; H3 is not yet testable and is deferred to the prospective version of the protocol. The prospective execution against the next MobilityData snapshot will be reported as a follow-up note in the same research programme.

This audit concerns the open specification itself, not the operators publishing under it. A2, A3, A6 and A7 detect patterns the GBFS specification *permits* (a constant placeholder, a conditionally-averaged virtual-station fleet, a zero-dock station, an empty capacity field). What our framework flags is that those permitted patterns are mutually incompatible with the arithmetic a downstream research consumer would naturally apply to the field. The technical lesson is about the standard, not about the publishers; criticism of the operators we name is out of scope for this paper. The standards-side recommendation in Section 5 is therefore phrased as a request for specification revision, not for operator sanction.

The audit is static: the `station_information` feed describes the deployment, not its dynamic behaviour. Real-time availability, rebalancing patterns and per-vehicle relocation live in the `station_status` feed, which carries its own quality issues and would require an analogous but disjoint audit protocol. The dynamic audit is the subject of the L2 limitation and the E4 future-work item.

5.3 Intellectual lineage in adjacent fields

The protocol’s design intent is consistent with an adjacent tradition of data audit that the mobility-data community has so far engaged with only sparingly.

Property-based testing in software engineering. Each of the seven anomaly classes A1–A7 admits a translation into a *property* that a compliant feed should satisfy: capacity is either a positive dock count, NaN or the same value across the system; coordinates lie within an a-priori perimeter; the $\bar{c}_{\text{profile}}/\bar{c}_{\text{actual}}$ ratio is bounded. This is the same intellectual move as in QuickCheck-style property-based testing for software, where invariants on the output are stated explicitly and checked across many inputs [47]. The Great Expectations and Pandera toolchains [17, 18] we benchmarked against are descendants of the same lineage. The contribution of the present audit, in that light, is to identify which properties GBFS implicitly assumes and to make their absence diagnosable; the SE community has spent two decades building infrastructure for that, but the mobility-data community has not yet imported it.

5.4 When a clean verdict hides a missing column

The Italian *Dott* case documented in Section 3.4 illustrates what is at stake: a country that passes A1–A6 cleanly is not necessarily clean: it may be silent on the only field that matters for infrastructure quantification. An audit framework that conflates *no problem detected* with *no data published* mis-reads the entire national corpus. A7’s empirical role is precisely to keep those two cases distinguishable.

5.5 GBFS v3 and the durability of the v2 legacy

GBFS v3 (released 2023) introduces `is_virtual_station`, a boolean that in principle disambiguates free-floating anchors from physical docks and would mechanically suppress A3 detection. The seven-class taxonomy nevertheless remains relevant under v3, for three reasons. Adoption is slow: of the 1,509 audited systems, only 541 (35.9%) declare GBFS v3 in their auto-discovery feed at the audit date, against 772 (51.2%) on v2.x; 50 remain on v1 and 146 expose no parseable version. The v2 legacy will persist: regulatory feeds on *transport.data.gouv.fr* and the MobilityData canonical catalogue archive snapshots that any longitudinal study from 2017 onward will need to ingest, and a v3 feed published today does not retro-fix the v2 historical record. And, most importantly, adoption is not compliance: the German cargo-bicycle mis-detection we report (Section 3.4) and the Italian *Dott* NaN-capacity propagation both occur on GBFS v3 feeds, because v3 makes the `is_virtual_station` flag *available* but does not make it *mandatory* or *verified*. The seven-class taxonomy A1–A7 therefore remains operationally relevant under full v3 adoption, and the protocol is straightforwardly extensible to the additional v3 attributes by adding rule-level checks at ingestion time.

5.6 Limitations

The protocol has five acknowledged limitations.

L1 – National coverage. The corpus is limited to metropolitan France; extension to overseas territories and other European jurisdictions requires revisiting thresholds, in particular the macro-regional surface threshold and the geofilter bounds.

L2 – Static versus dynamic. The present audit is deliberately scoped to the static `station_information` document. Auditing `station_status` (real-time availability) is a distinct research question that requires longitudinal data collection and a different analytical framework. The most consequential limitation that follows is the *zombie-station* risk: an operator that has physically dismantled a station but has not yet pruned the corresponding `station_information` entry will leave a ghost record that our static audit certifies as compliant, because the seven A1–A7 detectors all operate on the published schema. Downstream consumers of the Audit Catalogue who depend on station-presence should cross-check against a recent `station_status` snapshot, treating any entry whose status feed has been stale beyond a documented retention window as a zombie candidate. A 2-day pilot (experiment E4) already detected candidate dynamic anomalies on 12.9 % of monitored stations, and a fully operational collection and classification pipeline ships with the repository (`experiments/xp1_dynamic_audit/`); the formal dynamic extension is planned as a self-contained follow-up study.

L3 – Free-floating static versus dynamic scope. The seven-class taxonomy is applied to free-floating entries as much as to dock-based ones: every free-floating station carries an explicit `station_type = free_floating` label, and the A3 and A7 flags identify the operator-level capacity-publication anomalies that affect them (A3 for over-capacity profiles, A7 for NaN-capacity fields). The published parquet exposes both `capacity_raw` (the raw GBFS value, including placeholders and NaN) and `capacity_audited` (NaN for free-floating entries, dock-based count otherwise), so that downstream consumers have an unambiguous, audit-aware capacity to work with. What this v1.0 release does *not* audit is the *dynamic* behaviour of free-floating fleets – rebalancing patterns, real-time density of available vehicles, per-vehicle relocation – which would require an analogous audit of the `station_status` feed rather than `station_information`. That dynamic audit is out of scope here (it is the topic of L2); the present catalogue is therefore best characterised as a *static* audit of GBFS `station_information` across both dock-based and free-floating fleets, with the operational caveat that a free-floating station’s `capacity_audited` value is intentionally null.

L4 – Operator coverage. Some operators publish partial GBFS feeds or are unavailable during the audit window; the inventory completeness is therefore not guaranteed and must be re-evaluated periodically, which is the purpose of the temporal-stability experiment E3.

L5 – Scope and independence of the human check. The construct-validity check (Section 4.5) covers the rules verifiable from spatial imagery (A1, A3, A4, A5). A2 and A7 are structural data-property rules confirmed at the system level only (each is represented by a single operator in this catalogue snapshot), and A6 and the A3-boundary stratum had no eligible stations, so their per-station recall is not estimated. Most importantly, the check is *not independent of the authors*: the two annotators are this paper’s authors (domain experts from the same institution), so it establishes that the detectors track physical reality, not that they were validated by a disinterested third party. The blinding (to the pipeline and to each other) limits anchoring but cannot remove this conflict; an external, multi-institution annotation is the necessary next step and is left to future work. Four conventions, compatible with the current GBFS specification, would close most of the anomalies documented here: populate `vehicle_type` exhaustively (A1); leave `capacity` as null on virtual stations rather than using placeholders (A2); adopt the GBFS v3 `vehicle_status` extension for free-floating fleets and stop publishing per-vehicle anchors as `station_information` (A3); publish geographical perimeters as polygons in the `geofencing_zones` block (A4, A5); and avoid sentinel coordinates (0, 0) on stations whose geolocation is pending, instead omitting them from `station_information` until populated; a single such sentinel in Sevice Seville inflated its bounding-box area by four orders of magnitude in our E5 panel.

Methodological recommendation on A4 / A5 coupling. The Sevice finding (Table 11) suggests that the A5 bounding-box criterion be evaluated on the A4-filtered set of stations, not

on the raw set. We retain the as-shipped definition in this paper to make the dependency explicit, but downstream implementations should consider an A5 variant computed after the A4 outlier removal step.

5.7 Future directions

Each of the four limitations above is paired with a falsifiable experiment (Table 12); six of the nine include a completed pilot reported in this paper. The remaining priorities are the full four-dimensional threshold sweep of E2, the twelve-month temporal stability of E3, and the free-floating-native protocol of E6. Pilot scripts and the final experiment protocols will be packaged under `experiments/` of the source repository alongside the camera-ready release, so that the artefact can be challenged in public rather than defended in private.

Table 12: Nine falsifiable experiments mapped to the limitations of Section 5.6. “✓” = completed result reported in this paper; “Pilot ✓” = preliminary result already reported; full protocols and reproduction scripts ship under `experiments/` in the source repository.

ID	Title	Targets	Status
E1	Pre-registered held-out audit (post-freeze MobilityData)	Construct validity / circularity	Retrospective version
E2	Threshold and buffer sensitivity	$\sigma_{\max}$, $N_{\min}$, radii	$\sigma_{\max}$ pilot ✓
E3	Twelve-month temporal stability	Snapshot dependence (L4)	Running
E4	Dynamic extension to <code>station_status</code>	L2	2-day pilot ✓; Shanno
E5	European generalisation (live audit, 6 tech stacks)	L1	13-system panel ✓
E6	Free-floating-native protocol	L3	Planned 2027-Q2
E7	A4 v2 topology-aware ablation	Geospatial isotropy bias	✓ (Sec. 4.2)
E8	Leave-one-operator-out CV	Overfitting (H3)	✓ (Sec. 4.3)
E9	Dynamic <code>station_status</code> audit	Static/dynamic gap (L2)	Future work (pipel

5.8 Downstream use

The Audit Catalogue unlocks four families of downstream studies: supply-side composite-index work on cycling-environment quality at station and commune resolution; demand modelling with graph-based neural networks [48] whose false-station nodes from A3 are a known confound; spatial-equity analysis combining station access with INSEE Filosofi; and multimodal integration with GTFS, for which the hybridisation schema is compatible with the canonical GTFS validator [15].

Interactive dashboard. To make the audit verdicts directly inspectable without code, a focused Streamlit dashboard is shipped at <https://gbfs-audit.streamlit.app>, mirrored by `app/streamlit_app.py` in the source repository. The dashboard exposes five tabs: an overview with the corpus-level incidence of A1–A7; a row-level anomaly browser with live multi-flag filters and CSV export; an operator-level audit table with the per-class flag rates that drive the operator-driven hotspots reported in Section 3; a visual data dictionary of the 46 columns with completeness bars; and a free-text data explorer with a KD-tree-backed geographic map coloured by audit confidence. The dashboard is the artefact that turns the audit from a one-off paper into a continuously inspectable reference, addressing the persistence concern raised in Section 5.

Data and code availability

Lead contact. Requests for further information should be directed to and will be fulfilled by the lead contact, Rohan Fossé (rfosse@cesi.fr).

Materials availability. This study did not generate new physical materials. The released artefact is the *GBFS France Audit Catalogue* dataset, a derivative of public open data.

Data and code. The artefacts cover two scopes with explicit licensing boundaries.

- **French Audit Catalogue (Zenodo, ODbL v1.0).** The deposit contains four artefacts under the same concept DOI 10.5281/zenodo.20125460 (current release 1.0.0, 2026-05), mirrored on *Recherche Data Gouv*: (i) the 46,307-station certified Parquet (`stations_gold_standard_final.parquet`, ~ 6.6 MB, 46 typed columns including per-row anomaly flags and audited capacities); (ii) the lightweight system-level executive summary (`stations_gold_standard_audit_summary.csv`, ~ 60 kB, one row per audited system with anomaly counts and operator attribution) for users who want an overview before downloading the full parquet; (iii) the `rejected_stations.parquet` log with exclusion motives; and (iv) the audit report and machine-readable descriptors (JSON Schema, DCAT-AP, Frictionless Data Package, Croissant JSON-LD). The 142 candidate French GBFS feeds inventoried from *transport.data.gouv.fr* and MobilityData are archived with SHA-256 hashes in the same deposit; the certified subset of 123 systems is derived deterministically from them.
- **Source code (MIT licence).** The audit pipeline (`audit_pipeline` Python package, nine-step purging protocol, hybridisation joins, FAIR exporters) is released at <https://github.com/rohanfosse/gbfs-audit-catalogue>, together with the manuscript source, the companion Jupyter notebook (`notebooks/catalogue_recipes.ipynb`) and a focused interactive dashboard with five tabs (Overview, Anomaly browser, Operator audit, Schema, Data explorer) at `app/streamlit_app.py`, deployed publicly at <https://gbfs-audit.streamlit.app>. A CITATION.cff file and a Docker image (`ghcr.io/rohanfosse/gbfs-audit:1.0.0`) are provided for bit-exact reproduction. Releases are mirrored to Zenodo via the GitHub–Zenodo integration.
- **Hugging Face mirror.** The released parquet is mirrored as a Hugging Face dataset at <https://huggingface.co/datasets/rohanfosse/gbfs-audit-catalogue> for one-line access via `datasets.load_dataset()`.
- **Global audit snapshot.** The full per-system result of the 1,509-system MobilityData-catalogue sweep that underlies the global-projection section is frozen under `experiments/e2_threshold_sens`, the per-system `massive_audit_results.csv`, the aggregate `massive_audit_summary.json`, the `massive_audit.py` generator, and `verify_snapshot.py`, which re-derives and asserts the reported counts. Because the sweep crawls live operator feeds, a re-run drifts as the upstream catalogue evolves; the frozen snapshot is therefore the canonical, citeable artifact.
- **Additional resources.** The 500-station blinded sample used in the internal robustness checks, the orthogonal- and full-task classifier scripts, the $\sigma_{\max}$ sensitivity notebook, and the per-operator characterisation of the French free-floating subset are released as supplementary material alongside the dataset.

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CRediT authorship contribution statement

Rohan Fossé: Conceptualization, Methodology, Software, Data curation, Formal analysis, Validation, Visualization, Writing – original draft, Writing – review & editing. **Gaël Pallares:** Conceptualization, Methodology, Supervision, Project administration, Writing – review & editing.

Declaration of competing interests

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Declaration of generative AI and AI-assisted technologies in the writing process

During the preparation of this work the authors used GitHub Copilot and Claude (Anthropic) for code completion and editorial language polishing. After using these tools the authors reviewed and edited the content as needed and take full responsibility for the content of the publication.

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